End of Co-op Presentation





Presented by: Jonah Wolk

My Background

About Me

- From St. Louis, MO
- I have a dog
- Three siblings
- Ambidextrous-ish
- Music

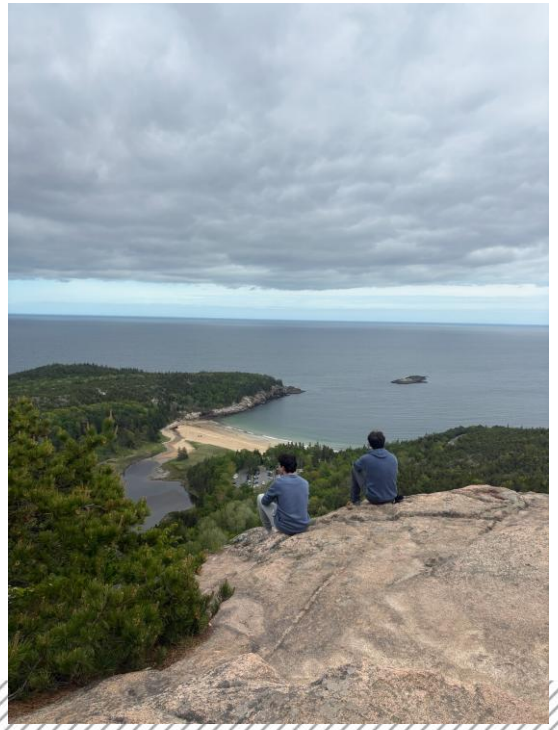






What led me to WSP

- Attending Northeastern University (Boston)
- Getting a BSME
- Wanted to gain valuable work experience in a first-rate engineering firm
- Maine



LVX VERITAS VIRTVS

Northeastern University

Experience Before WSP

- Lifeguarding
- Counseling
  - Interpersonal Skills & Supervision-Based Environment
- Theme Park Engineering & Ecoscholars
- Intro2Engineering
- College Courses
  - Interactive Energy Sim.
  - MATLab/Fusion360/SolidWorks
  - Soldering, Circuits, & Other Technical Work
- Ikea Furniture







Interactive Energy Simulator

- Kinetic -> Electrical
- Youth-based Audience
- Color Sensor finds
  RPM -> LED Array
- Teaches Optimization



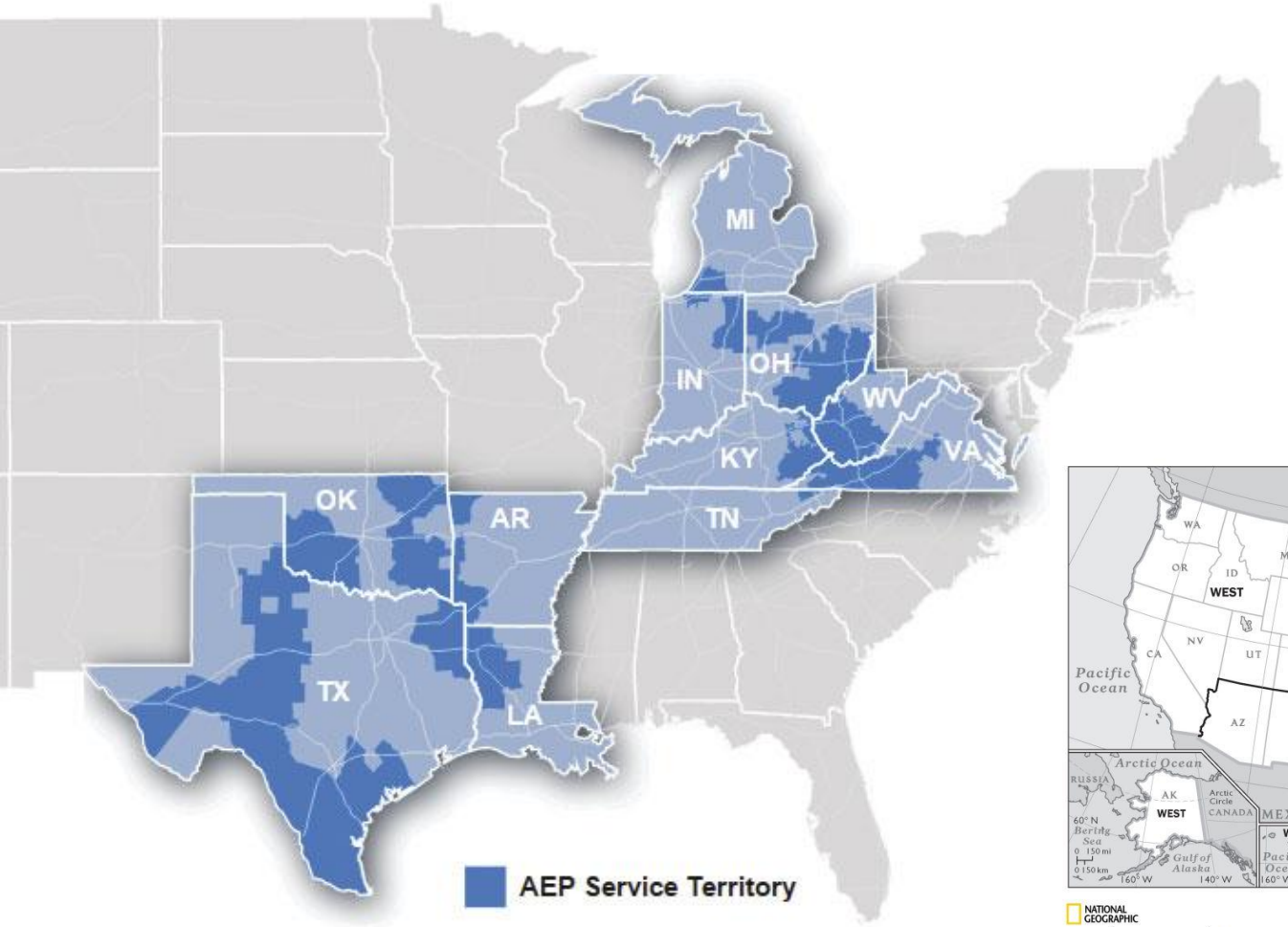


Relevant Course Work

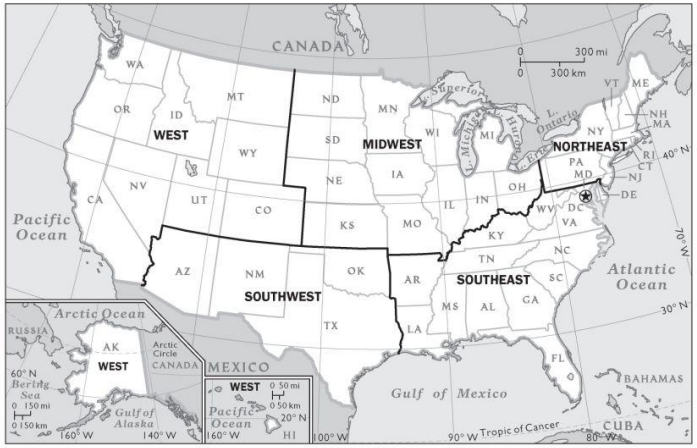
- Statics
- Physics
- Material Science
- Thermodynamics
- Calculus
- Engineering Cornerstone
- Chemistry/Biology
- Various Labs

Projects

AEP



- Southwest
- Southeast
- East Midwest



NATIONAL GEOGRAPHIC education

UNITED STATES REGIONS

TTMP – Dyess 3 to Dyess 1

- Outskirts of Abilene, Texas
- Started from a DTLM Line Edit
- Completed:
  - General Model Setup (WCs/Zones)
  - IKE Data Translation
  - Attached ADSS
  - Matched Tension/LiDAR
  - Structure Analysis & Misc. Reports



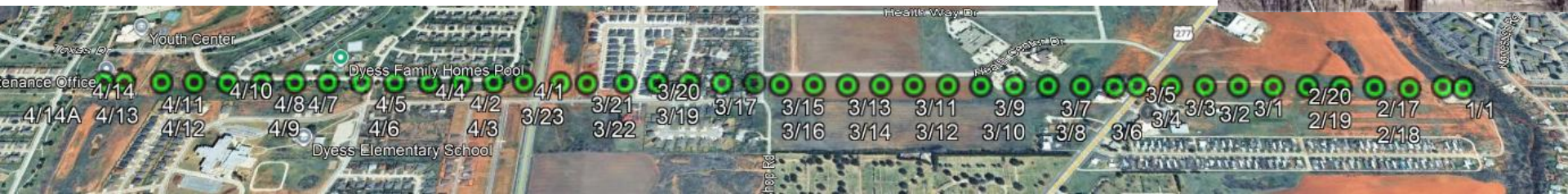
TTMP – Dyess 1 to S. Abilene

- More of the same as Dyess 1-3
- Connecting Line





- Distribution Underbuilds
- More Guys/Stub Poles
- 2x Structure Count

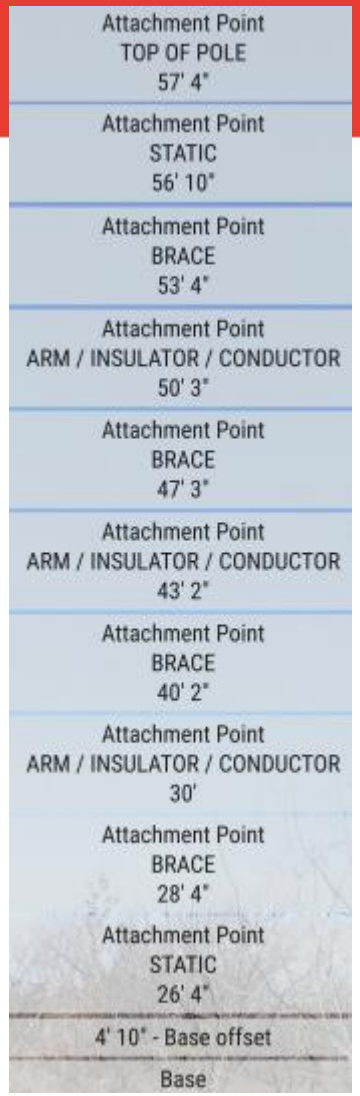


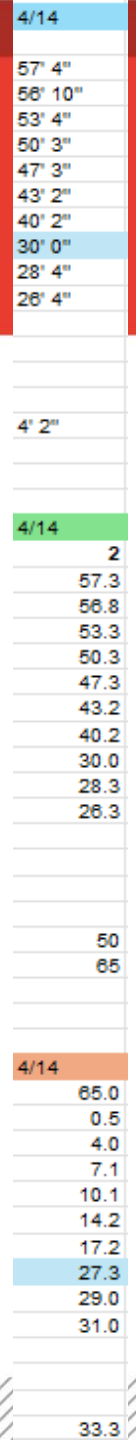
What I Learned



"If you're doing something on excel, there's probably a easier way to do it"
-Ken Drelich 2026

- There is ALWAYS a way to make your life easier
- Basic Pole Modeling
- PLS CADD Proficiency
- Creating & Stringing via Attachments
- Setting up a Greenbook





Challenges

- DTLM -> More Precise
- Old Structures (60s/80s)
  - Illegible Birthmarks
  - Outdated Techniques
- New Excels
  - Structural Analysis & Clearance Analysis
- Tensioning & Matching LiDAR
- Non-Converging Structures

Del Sol - Frontera

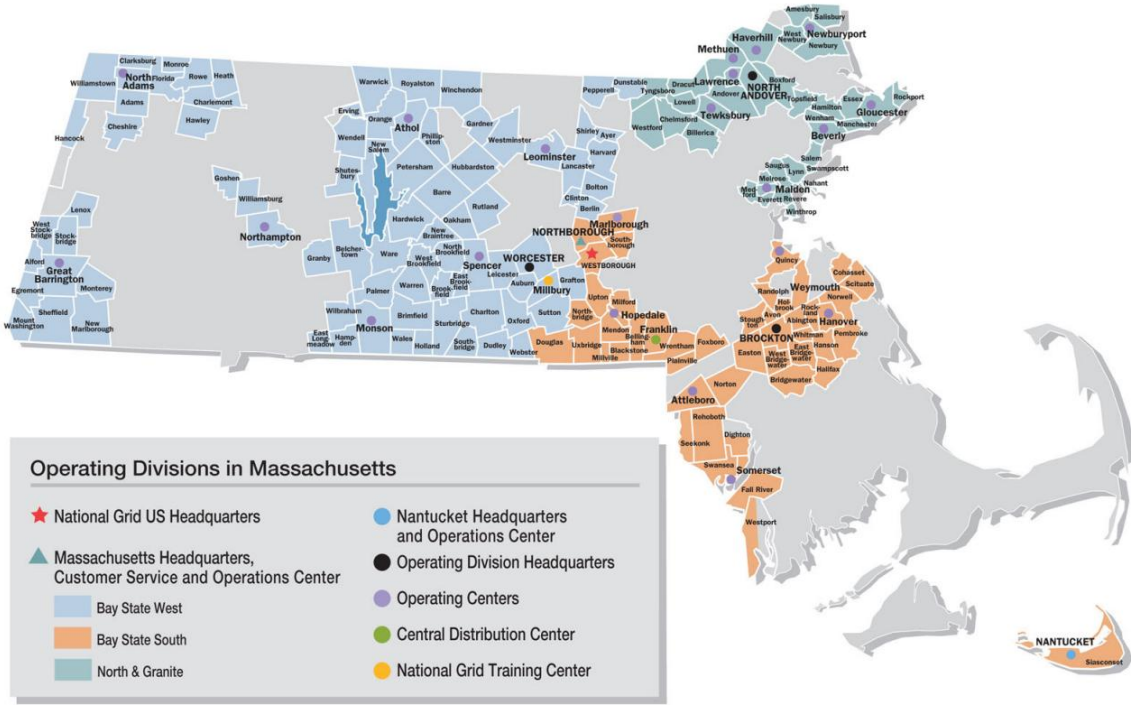
- AEP
- Southern Texas (near Rio Grande City)
- 36 miles, 345kV
- First Task

- Ran Reports
  - Stringing, Structure Clearance, Usage, etc
- Structure Shift (x2)
  - Autosagging, PI Shifts, Wire/Wire Clearances



National Grid

nationalgrid



-Central & Western Mass

E5/F6 MFAD

-NatGrid Project

-67 Miles, 711 Structures

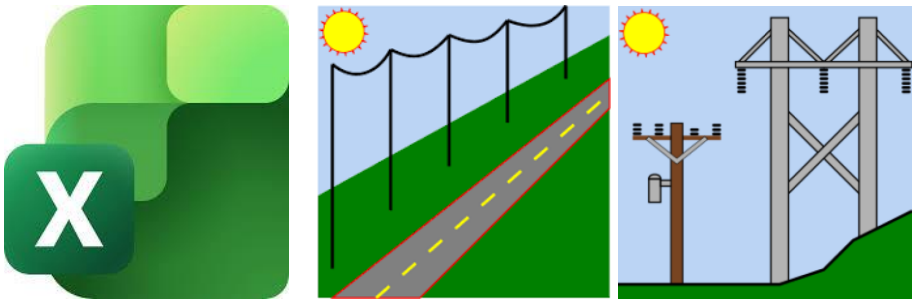
-69kV Lattice -> 115kV Steel Monopoles

- GPR Review
- MFAD Input Spreadsheet
  - Boring Data
  - Soil Data
  - Case Associations

Takeaways

Technical Skills

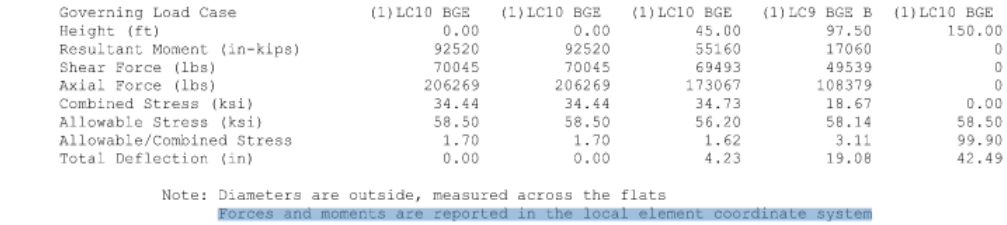
- Modeling structures with PLS Pole
- Excel Experience
- Conductors/Insulators/Vangs & their Use Cases
- QAQC (Markups & Checks)
- Different Client Procedures (AEP vs. BGE vs. Xcel)
- Running reports & Setting up Packages



Other Skills

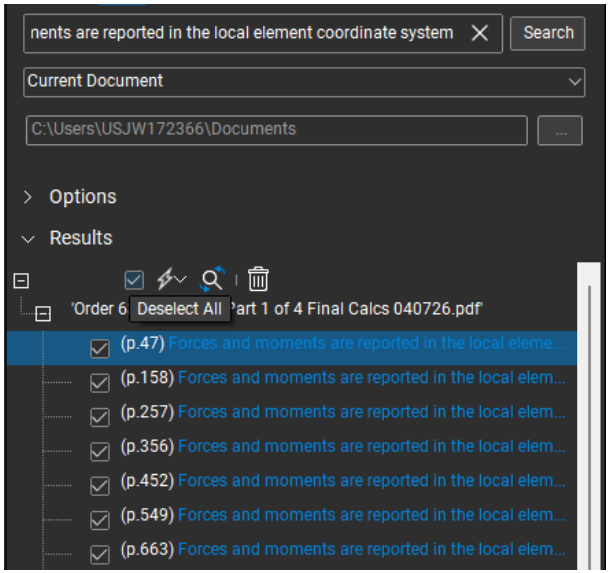
Bluebeam Revu

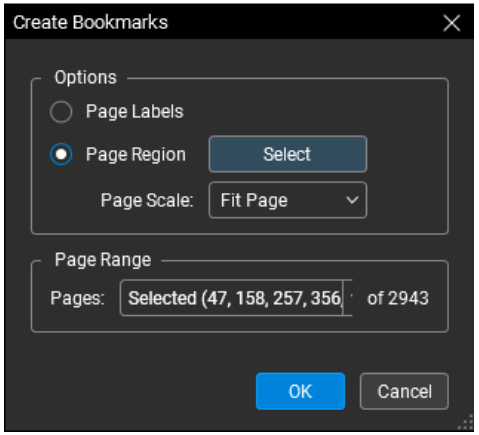
1. Ctrl F
2. Ctrl C, Select All
3. Create Bookmarks
4. Page Region ->
   a) Select
   b) Highlight Str. Name
   c) OK



Governing Load Case	(1)LC10 BGE	(1)LC10 BGE	(1)LC10 BGE	(1)LC9 BGE B	(1)LC10 BGE
Height (ft)	0.00	0.00	45.00	97.50	150.00
Resultant Moment (in-kips)	92520	92520	55160	17060	0
Shear Force (lbs)	70045	70045	69493	49539	0
Axial Force (lbs)	206269	206269	173067	108379	0
Combined Stress (ksi)	34.44	34.44	34.73	18.67	0.00
Allowable Stress (ksi)	58.50	58.50	56.20	58.14	58.50
Allowable/Combined Stress	1.70	1.70	1.62	3.11	99.90
Total Deflection (in)	0.00	0.00	4.23	19.08	42.49

Note: Diameters are outside, measured across the flats
Forces and moments are reported in the local element coordinate system





BY VALMONT INDUSTRIES FOR:
Design Id: A-2047-W

Other Applications:

- Mass exporting similar dwgs/calc sheets
- Highlighting/adding markups
- Exporting & inputting text into excel (names/data)

Overall Takeaways

- Work Culture & Professionalism
- OK to ask for help
- Setting up Check Packages
- Excel, Bluebeam, & ProjectWise Proficiency
- Modeling Best Practices
  - Saving Frequently
  - Line Edits (Save a Default)



What I enjoyed

- Expanding my Excel/Bluebeam skills
- Mentor/Intern Meetings
- Getting to know & work with everybody
- Freeport/Maine (Acadia)
- Brown Bags & Lunches
- Experience!
- “Hanging out with Sam”

  - Sam Belliveau 2026



Any Questions?